

Meet Patel

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EDUCATION

MS in Computer Information System | University of Houston ClearLake, Houston, TX | GPA : 3.88 Jan 2024 – Dec 2025
Courses : Data warehouse & BI, Artificial Intelligence, Big Data, Machine Learning, Predictive Analysis, Advance OS

Bachelor of Computer Application | Charutar Vidhya Mandal University, Gujarat, IN | GPA : 4.00 Mar 2020 – May 2023
Courses : DBMS,SQL,Python,R,Data Mining, Data Structures & Algorithms, Statistics & Probability

PROJECTS

CSCI - 6838 Research Project and Seminar – Graduate Capstone Project ([Capstone Project](#)) August 2025 to Present

Role : Technical Lead I (Backend Development & Data Handling, Integration)

- Led the design and development of an **Agentic AI application** that automates covered-call option screening and recommendations for retail investors.
- Addressed the challenge of **non-deterministic AI behavior** in financial decision systems by implementing **Test-Driven Development (TDD)** and reproducible workflows using **LangGraph & LangChain** Agentic AI frameworks.
- Engineered a **multi-agent architecture** integrating real-time market data from **Polygon.io** and **Alpha Vantage APIs**, enabling accurate analysis of strike, premium, delta, and expiration parameters.
- Built a modular system with an **Options Data Agent** for market retrieval and a **Strategy Agent** for AI-driven ranking of contracts based on ROI and risk exposure and multi-trade opportunities.
- Delivered an interactive **web interface** for intuitive visualization of portfolio insights and option recommendations.
- Improved investor decision-making efficiency by automating data aggregation and strategy evaluation, demonstrating measurable impact in financial workflow automation
- Collaborated with mentors and faculty to ensure adherence to **software engineering best practices, SDLC documentation, and unit-test-verified AI consistency.**

Tech Stack : Python, LangGraph, LangChain, Streamlit, Polygon API, Alpha Vantage, OpenAI GPT-5, Git, TDD, SDLC, REST APIs

Custom Linear Regression for Housing Price Prediction May - 2025

- Implemented a from-scratch **Linear Regression model** with **Gradient Descent** in Python to predict California housing prices, which removed dependencies on external ML libraries and deepened understanding of core optimization mechanics.
- Curated and ingested the California Housing dataset (via scikit-learn or local CSV) and applied manual feature scaling, enabling stable gradient updates; **this led to** faster and more reliable convergence.
- Achieved highly accurate predictions with an average absolute difference of approx. \$349 and a maximum difference of around \$1,000 compared to scikit-learn's model, demonstrating strong model performance.
- Designed a training and evaluation workflow that reports MAE and MSE, which resulted in clear, quantifiable baselines and enabled targeted error analysis.

Tech Stack : Python 3, Jupyter Notebook, NumPy (from-scratch Linear Regression + Gradient Descent)

U.S. COVID-19 Vaccination Analysis at County Level March 2025 - Present

- Led a comprehensive analysis of COVID-19 vaccination disparities across U.S. counties using Python, pandas, NumPy, seaborn, and matplotlib, uncovering critical urban-rural and demographic trends.
- Merged and cleaned large datasets from the CDC and U.S. Census Bureau, standardizing FIPS codes and demographic data to ensure data integrity.
- Visualized geospatial and time-series insights via choropleth maps and interactive plots (Matplotlib/Seaborn/Plotly) to support data-driven decisions.
- Automated repeatable analyses and data quality checks with modular notebooks and functions, improving maintainability and reproducibility.

Tech Stack : Python 3.9+, Jupyter Notebook & Google Colab, pandas, NumPy, Matplotlib, Seaborn

Secure Distributed Service Broker System January - 2025

- Designed and implemented a secure distributed system using Python 3.x, TCP Sockets, and Tkinter, featuring a service broker and multiple providers for seamless service discovery and interaction.
- Implemented a multithreaded server with Python threading to handle concurrent client connections reliably, improving responsiveness underload and ensuring thread-safe request handling.
- Implemented robust security measures, including Secured service access with ID/password authentication and cryptographic hashing using hashlib (MD5 through SHA-512), enforcing credential verification and strengthening data integrity.
- Developed a user-friendly Tkinter-based client GUI, enhancing user experience and facilitating easy interaction with the system.

Tech Stack : Python 3, TCP sockets, threading, hashlib (MD5/SHA-1/SHA-256/SHA-512), Tkinter, Hash Algorithm

SKILLS

- Programming & Analytics:** Python (3.9+ | Pandas | NumPy | Scikit-learn | Matplotlib | Seaborn), SQL
- AI & Agentic Frameworks:** LangGraph, LangChain, OpenAI GPT Models, Google Gemini APIs
- Data Visualization:** Power BI, Tableau, Excel (Pivot Tables, VBA, X-Lookup)
- Databases & Cloud:** MySQL, RDBMS, MongoDB, Snowflake
- ML & Modeling:** Regression, Classification, Clustering, Time-Series Forecasting, Option Pricing Models